English Medium

EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

Topic wise Notes

LECTURE 30

Animal Diversity

Animal Diversity

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Animal Diversity refers to the variety of animals present on Earth based on their body organization, symmetry, body cavity, segmentation, and evolutionary relationships. Animals are classified into different phyla according to their characteristics.

All Eukaryotic, multicellular hetrotroph, without cell wall organisms are kept.

Note : Unicellular and Eukaryotic animal are kept in Protista kingdom according to R.H. whittekar. But unicellular animals are studied in animalia kingdom.

Oldest classification of Animalia was given by Aristotle he classified animal into two categories:

Anaima— Animals lacking red blood. **E$_g$:** Porifera coelentrata to Aschelminaties.

Enaima— Animals having red blood in their body cavity. **E$_g$:** Annelida, Echinode, mata, chordata, mollasca.

Branch of science deals with study of animal known or zoology.

Father of zoology–Aristotle

Book-Historic animalia.

Animalia		
Protozoa Unicellular animal	**Metozoa (Multicellular animal)**	
	Non-Chordata	**Chordata**
	Notochord absent	Notochord present
	Porifera – Annelida	Pisces
	Coelenterate - Arthropoda	Amphibian
	Ctenophora- Mollusca	Reptile
	Platyhelminthes- Echinodermata	Aves
	Aschelminthes- Hemichordata	Mammals

Level of Organisation

Protoplasmic –Protozoa (Amoeba)

Multicellular- Porifera (Sponges)

Tissue – Coelentrata (Hydra) & Ctenophore (Ctenophora)

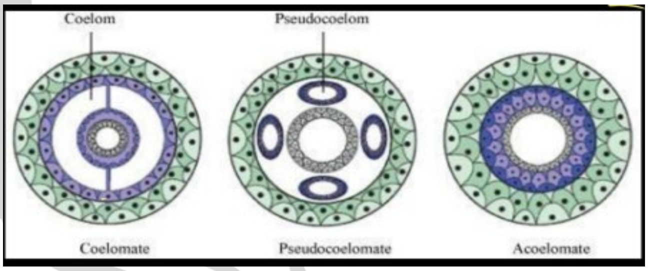
Organ – Platyhelminthes (Flat worm Planaria)

Organ system – Aschelminthes, Annelida, Arthropoda, Mollusca, Echinodermata Chordata.

Symmetry		
Asymmetrical	**Radial**	**Bilateral**
Body cannot be divided into two equal halves.	Body can be divided into more than two equal halves.	Body can be divided into only two equal halves.
E$_g$: Porifera (sponges)	**E$_g$:** Ctenophora echinodermota (star fish) Mullasca	**E$_g$:** Annelida Arturopoda Churdata.

Body Cavity

The term **"Body Cavity"** appears to be a common misspelling of Body Cavity. A body cavity is a fluid-filled space within the body that houses and protects internal organs. These spaces allow organs like the heart and lungs to expand and contract without interfering with other tissues



Germinal layer	
Diploblastic	**Triploblastic**
Two germinal layers	Three germinal layers
1. Ectoderm 2. Endoderm **E.g.** Coelenterate Ctenophora	1. Ectoderm 2. Endoderm 3. Mesoderm **E.g:** Platyhelminthes to Chordata

Basis of Classification:

Levels of Organisation:

- **Organ Level:** Tissues group to form organs (e.g., Platyhelminthes).

- **Organ System Level:** Organs form systems for specific functions (**Eg,** Aschelminthes to Chordata).
- **Cellular Level:** Cells arranged as loose aggregates (e.g., Porifera).
- **Tissue Level:** Cells perform the same function form tissues (**Eg** Coelenterate, Ctenophora)

Symmetry:

- **Asymmetry:** No plane divides the body into equal halves (e.g., most sponges).
- **Radial Symmetry:** Any plane through the central axis divides the body into identical halves (e.g., Coelenterates, Ctenophores, adult Echinoderms).
- **Bilateral Symmetry:** Only one plane divides the body into identical left and right halves (e.g., Annelids, Arthropods, Chordates).

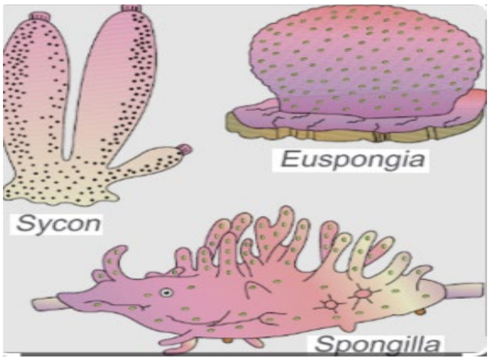
Germinal Layers:

- **Diploblastic:** Two embryonic layers (ectoderm and endoderm) (e.g., Coelenterates).
- **Triploblastic:** Three embryonic layers (ectoderm, mesoderm, and endoderm) (e.g., Platyhelminthes to Chordata).

Coelom (Body Cavity):

- **Acoelomates:** Body cavity absent (e.g., Platyhelminthes).
- **Pseudocoelomates:** Body cavity not lined by mesoderm, present as scattered pouches (e.g., Aschelminthes or roundworms).
- **Coelomates:** Possess a true coelom, a body cavity lined by mesoderm (e.g., Annelids to Chordates).
- **Notochord:** A mesodermally derived rod-like structure on the dorsal side during embryonic development. Animals with it are chordates; those without are non-chordates (Porifera to Echinodermata).

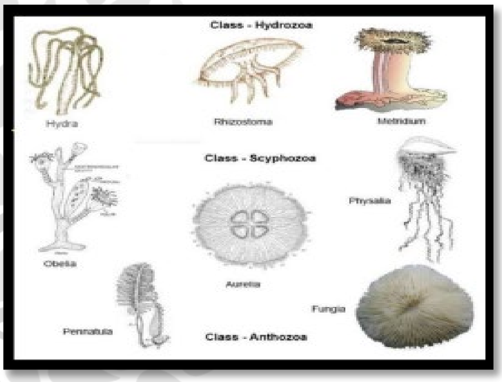
Porifera

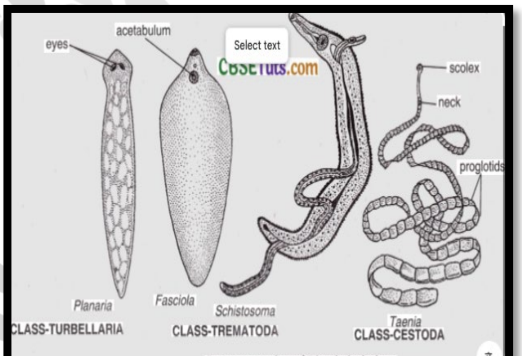


- Multicellular animal
- Tissue absents
- Asymmetrical
- Porec are present in the whole-body surface known as 'ostia'.
- **Eg :** Sycon Euspongia Spongilla sponges.

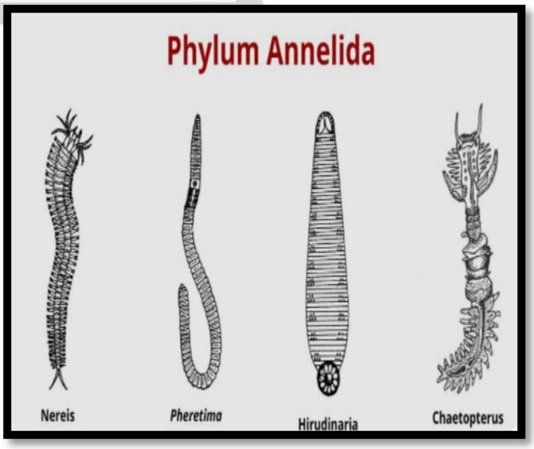
Coelenterate or Cnidaria

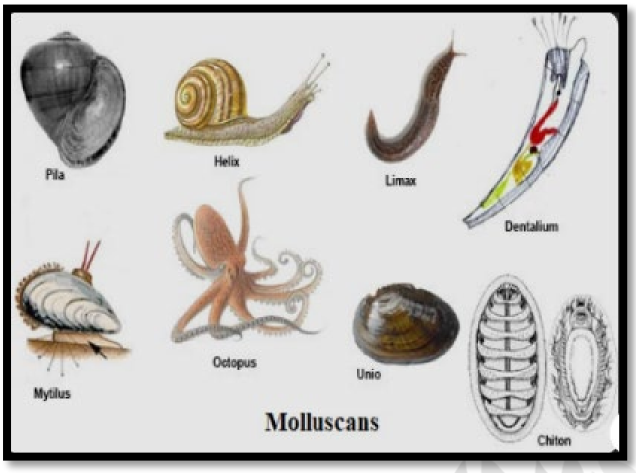
- Tissue present
- Diploblastic
- Acelomate
- Radia
- Cnidoblast cells are present for its sensitivity
- **Eg:** Hydra amelia (jelly fish) Ademsia (Sea anemone) mango obelia physalia.

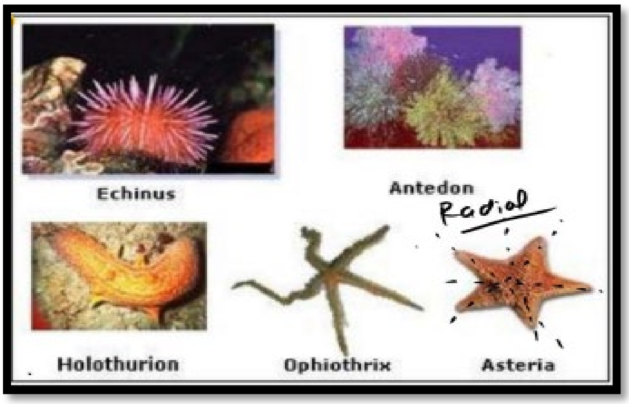


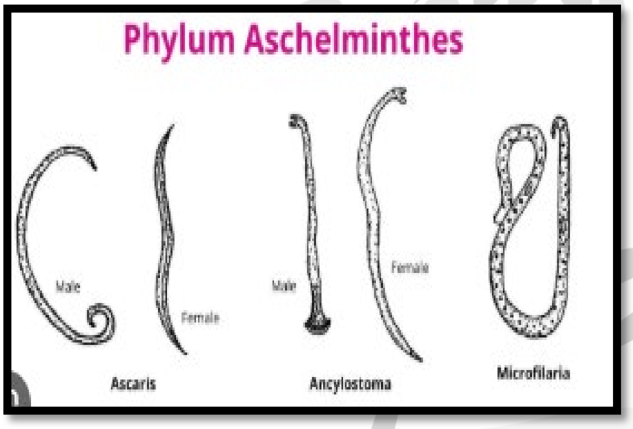


Some examples of Platyhelminthes.



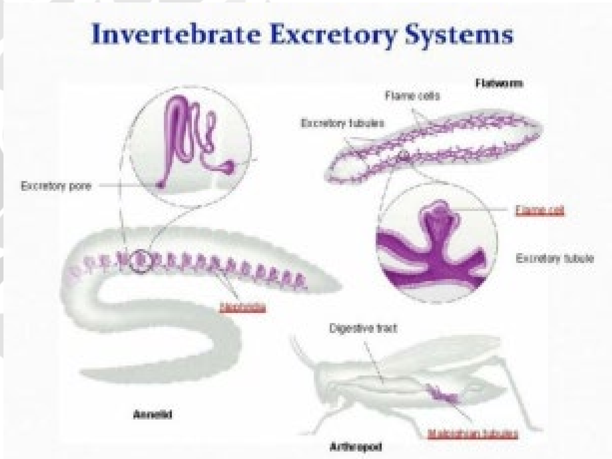






Key Phyla Characteristics (Non-Chordates):

- Phylum Coelenterata (Cnidaria): Aquatic, tissue level, radial symmetry, stinging cells (cnidoblasts), gastro-vascular cavity, existence in polyp and/or medusa forms, some show metagenesis (alternation of generations) (**E**$_g$ Hydra, jellyfish)
- **Phylum Ctenophora (Comb Jellies):** Exclusively marine, radial symmetry, locomotion via 8 comb plates, bioluminescence is well-marked. Phylum
- **Platyhelminthes (Flatworms):** Bilateral symmetry, triploblastic, acoelomate, organ level, dorso-ventrally flattened body, flame cells for excretion/osmoregulation, many are endoparasites.
- **Phylum Aschelminthes (Roundworms):** Bilateral, triploblastic, pseudocoelomate, organ system level (e.g., Ascaris).
- **Phylum Annelida (Segmented Worms):** Bilateral, triploblastic, coelomate, organ system level, metameric segmentation, closed circulatory system, nephridia for excretion (**E**$_g$ earthworm, leech).



- **Phylum Arthropoda:** Largest phylum. Bilateral, triploblastic, coelomate, organ system level, chitinous exoskeleton, jointed appendages, open circulatory system, respiration via gills, book gills, book lungs, or tracheae (e.g., insects, Follow Us O f spiders, crabs).
- **Phylum Mollusca:** Second largest phylum. Bilateral, triploblastic, coelomate, organ system level, body with head, visceral hump, and muscular foot, usually a shell present, a file-like rasping organ called radula for feeding.
- **Phylum Echinodermata (Spiny-skinned):** Marine, organ system level, adults with radial symmetry but larvae with bilateral symmetry, endoskeleton of calcareous ossicles, presence of a unique water vascular system for locomotion and food capture.
- **Phylum Hemichordata:** Worm-like marine animals, an anterior proboscis, collar, and trunk,

open circulatory system, proboscis gland for excretion.

Characterized by a notochord, dorsal hollow nerve cord, and paired pharyngeal gill slits.

- **Subphylum Vertebrata:** Possess a vertebral column (backbone).
  - **Class Pisces (Fish):** Aquatic, 2-chambered heart, gills for respiration,
  - **Class Amphibia:** Semi-aquatic/terrestrial, moist skin without scales, 3-chambered heart (except crocodile which has 4), require water for reproduction, larva (tadpole) metamorphoses.
  - **Class Reptilia:** Predominantly terrestrial, dry skin with scales, 3-chambered heart (4 in crocodiles), internal fertilization, oviparous (egg-laying).

<table>
<tr><th colspan="3">Chordates</th></tr>
<tr><th>Non-vertebreta</th><th colspan="2">Vertebrata</th></tr>
<tr><td>Notochord remains present whole life.</td><td colspan="2">Notochord later replaced by vertebral column</td></tr>
<tr><td rowspan="2">Agantha
E.g:
Cyclostomata
Sea organism</td><th colspan="2">Pisces</th></tr>
<tr><td>Chondirchyles</td><td>Amphibians ,Reptile ,Aves mammals</td></tr>
</table>

Class Aves (Birds): Feathered, forelimbs modified into wings, pneumatic (hollow) bones, 4-chambered heart, homeothermic (warm-blooded), air sacs connected to lungs.

Class Mammalia: Mostly terrestrial, presence of hair, mammary glands, 4- chambered heart, homeothermic, viviparous (except platypus).